Systems of Linear Equations — Answer Key

Type 1 Is It a Solution?

1. $2+3=5$ ✓; $4-3=1$ ✓

Answer: Yes

2. $3\cdot 1+1=4$ ✓; $1+4=5\neq 6$ ✗

Answer: No

3. $-2+6=4$ ✓; $-2\cdot(-1)=2$ ✓

Answer: Yes

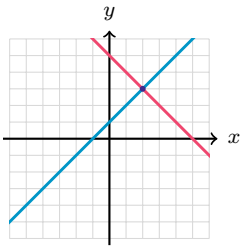
4. $3-(-2)=5$ ✓; $3+2(-2)=-1\neq 1$ ✗

Answer: No

Type 2 Solve by Graphing

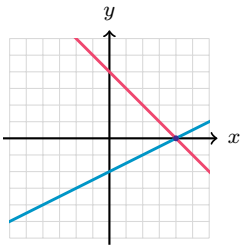
5. Graph both lines; they cross at $(2,3)$. Check it in both equations.

Answer: $(2,3)$



6. Graph both lines; they cross at $(4,0)$. Check it in both equations.

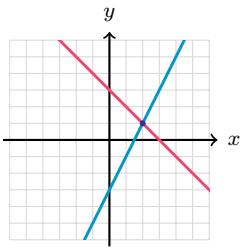
Answer: $(4,0)$



© 2026 FangMath. All rights reserved.

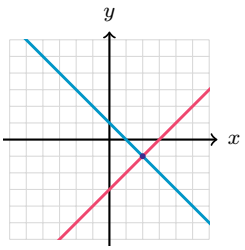
7. Graph both lines; they cross at $(2, 1)$. Check it in both equations.

Answer: $(2, 1)$



8. Graph both lines; they cross at $(2, -1)$. Check it in both equations.

Answer: $(2, -1)$



Type 3 Substitution

9. $x + 2x = 12,\ 3x = 12,\ x = 4;\ y = 2 \cdot 4 = 8$

Answer: $(4, 8)$

10. $2x + x + 3 = 9,\ 3x = 6,\ x = 2;\ y = 5$

Answer: $(2, 5)$

11. $3(y - 1) + y = 13,\ 4y = 16,\ y = 4;\ x = 3$

Answer: $(3, 4)$

12. $x - (-3x + 2) = 6,\ 4x = 8,\ x = 2;\ y = -4$

Answer: $(2, -4)$

13. $4x - 5 = 2x + 1,\ 2x = 6,\ x = 3;\ y = 7$

Answer: $(3, 7)$

© 2026 FangMath. All rights reserved.

14. $3y - 8 + 2y = 7,\ 5y = 15,\ y = 3;\ x = 1$
Answer: $(1, 3)$

Type 4 Elimination

15. Add: $2x = 14,\ x = 7;\ y = 3$
Answer: $(7, 3)$

16. Subtract: $4y = 12,\ y = 3;\ 2x = 4,\ x = 2$
Answer: $(2, 3)$

17. Add: $4x = 12,\ x = 3;\ y = 2$
Answer: $(3, 2)$

18. Multiply the first by 2: $2x + 4y = 16$. Subtract: $x = 4;\ y = 2$
Answer: $(4, 2)$

19. Multiply the first by 2: $4x + 2y = 14$. Add: $7x = 21,\ x = 3;\ y = 1$
Answer: $(3, 1)$

20. Multiply the second by 3: $6x - 3y = 9$. Add: $10x = 10,\ x = 1;\ y = -1$
Answer: $(1, -1)$

Type 5 How Many Solutions?

21. Same slope 2, different y-intercepts: parallel lines.
Answer: No solution

22. Different slopes (3 and -1): the lines cross once.
Answer: One solution, $(2, 4)$

© 2026 FangMath. All rights reserved.

23. Divide the second by 2: $y = \frac{1}{2}x + 3$. Same line.

Answer: Infinitely many

24. Same slope -1, different intercepts (5 and 9).

Answer: No solution

25. The second is the first times 2: the same line.

Answer: Infinitely many

26. Slopes -4 and 4 are different: one crossing, at $(0, 1)$.

Answer: One solution, $(0, 1)$

Type 6 Word Problems

27. $x + y = 20,\ x - y = 6$. Add: $2x = 26$.

Answer: 13 and 7

28. $c + w = 15,\ 2c + 4w = 44.\ 2c + 2w = 30$, subtract: $2w = 14$.

Answer: 8 chickens, 7 cows

29. $p + n = 10,\ 2p + 3n = 24.\ 2p + 2n = 20$, subtract: $n = 4$.

Answer: 4 notebooks (and 6 pens)

30. $C = 10m + 30$ and $C = 15m + 15$: $10m + 30 = 15m + 15,\ 15 = 5m$.

Answer: 3 months, \$60

© 2026 FangMath. All rights reserved.